

Role Support in Knowledge-Graph Error Ranking: Predictor Regimes and Evaluation Policies

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First define what counts as an error

The study scores recovery of recorded held-out answers.

Error = 1 if predicted tail is absent from recorded targets

- T is the set of recorded held-out answers for one query.
- An unrecorded true answer can still count as an observed-target error.
- Features use the predicted tail; hidden answers cannot enter the decision.

Counts and missing-role bits are different representations

A zero-count indicator is determined by its count; it adds no new information.

Head-role count

$n_H(h, r)$ counts training triples where h appears as the head of relation r .

Tail-role count

$n_T(\hat{t}, r)$ counts training triples where the predicted tail appears in the tail role of relation r .

Missing-role bits

$b_H = 1[n_H = 0]$ and $b_T = 1[n_T = 0]$. An explicit bit can change a fitted model without adding conditional information.

- The raw score $U = (b_H + b_T) / 2$ penalizes both missing roles equally.
- An incomplete graph cannot establish falsehood from training absence.

An audit across predictors and scorers

Every score comparison conditions on a defined prediction population.

Component	Executed scope
Static graphs	FB15k-237; WN18RR
From-scratch predictors	12 DistMult / ComplEx runs
Continued training	6 warm-start ComplEx stages
Ensembles	6 fixed three-member ensembles

- Original test populations: 10,301 and 2,689 grouped queries.
- Shared graphs, seeds, and warm starts are not independent replications.

The raw missing-support score reverses direction

Across two FB15k-237 training regimes, higher accuracy accompanies a reversal in raw-score ranking.

FB15k-237 regime	Accuracy	Missing-role AUROC
Original ComplEx	10.05%	0.6893
Continued ComplEx	44.35%	0.3639
Direction reference	—	0.5 = chance ranking

- Higher U means more missing support; AUROC below 0.5 reverses that intended error ordering.
- Training changes occur together; accuracy is not isolated as the cause.

Learned counts can still help beyond disagreement

Selected fixed-ensemble comparisons: adding continuous role counts reduces AURC.

Ensemble / dataset	LR Δ AURC	HGB Δ AURC
DistMult / FB15k-237	-0.071931	-0.032414
ComplEx / FB15k-237	-0.070314	-0.035930
Continued / FB15k-237	-0.008947	-0.004599
Continued / WN18RR	-0.000438	-0.000932

- LR = logistic regression; HGB = histogram gradient boosting. Comparator includes confidence, frequencies, and disagreement.
- Counts improve AURC and Brier in all 12 ensemble/head comparisons; operating-point dominance does not follow.

High-coverage results can run out of headroom

The best possible risk depends on how many correct predictions the fixed model already made.

Hindsight floor

If c predictions are correct and k answers are retained, minimum possible error is $\max(0, 1 - c/k)$. It uses correctness after the fact.

Original runs

Counts logistic regression reaches the 85%-answer-rate floor in all 12 original predictor runs.

Continued runs

It reaches the floor in none of the 6 continued-training runs, although remaining high-coverage headroom is small.

- A hindsight floor explains an endpoint; it is not a deployable selector.
- Binary additions can help one metric and worsen another.

Candidate restrictions can help one graph and harm another

Restricting tails to training-supported roles changes the predictor itself.

Original DistMult	FB15k-237	WN18RR
Original accuracy	10.56%	38.16%
Restricted accuracy	25.79%	28.05%
All recorded answers excluded	6.63%	27.67%

- All test queries stay in the denominator, including unreachable targets.
- Observed role support is not a hard truth constraint.

A fixed threshold changes the decision being evaluated

A development rule targeting 85% answers does not necessarily answer 85% on test.

DistMult policy	Test answer rate	Test error ratio
FB15k-237: base LR	87.41%	88.40%
FB15k-237: + binary	82.86%	87.33%
WN18RR: base LR	82.97%	55.18%
WN18RR: + binary	83.67%	55.30%

- Lower error may accompany fewer answers; compare achieved rates.
- Reported error ratio uses expected error and acceptance counts under boundary randomization.

Support as a predictor-specific diagnostic

Role support is a conditional feature to test, not an invariant uncertainty label.

- Freeze the predictor, candidates, target definition, and information available at decision time.
- Compare continuous counts against confidence, frequencies, and disagreement.
- Report ranking, probability scoring, and achieved answer rates separately.
- Test new graphs and temporal transfer before making deployment claims.

Source: github.com/ChorokLeeDev/kg-uncertainty-revision